

Highlights

A Reproducible Methodological Framework for Prosecutorial Congestion Risk Prediction Using Explainable Machine Learning and Temporal Validation

- Congestion risk at the prosecutorial stage predicted from open national records
- Target-defining variables excluded; all thresholds learned on training years only
- LightGBM reaches F1 0.469 and ROC-AUC 0.798 on the 2025 hold-out year
- Ablation and SHAP agree that the risk signal is mainly territorial-institutional
- Scores rank combinations for review but are overconfident at mid-to-high risk

A Reproducible Methodological Framework for Prosecutorial Congestion Risk Prediction Using Explainable Machine Learning and Temporal Validation

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ABSTRACT

Case congestion in public prosecution services delays responses to citizens and complicates the allocation of staff and resources, yet predictive studies of justice systems have concentrated on courts. This study estimates a proxy risk of prosecutorial congestion from the open administrative records of the Public Prosecutor's Office of Peru (2019–2026; 9,593 records). The target is defined by percentile rules on case backlog and case processing rate, with thresholds learned on the training years only and examined in a sensitivity analysis. The variables that define the target were excluded from the predictors, features were selected by the consensus of six methods, and nine model configurations were compared under year-block walk-forward validation, with model and threshold selection on 2024, a hold-out test on 2025, and a prospective evaluation on the partial 2026 data. The selected LightGBM model reached an F1-score of 0.469 (95% bootstrap confidence interval 0.390–0.546) and a ROC-AUC of 0.798 on the 2025 test set, and an F1-score of 0.559 and a ROC-AUC of 0.866 on the 2026 data. An ablation study and SHAP attributions agree that the signal is mainly structural: territorial and institutional features carry most of the predictive information, whereas lagged historical features add little. Because the scores are overconfident in the medium-to-high range, they are better suited to ranking operational combinations for preventive review than to serving as calibrated probabilities. The study provides a leakage-controlled, temporally validated reference for prosecutorial congestion analytics and delimits where its use is warranted.

1. Introduction

Congestion in justice systems increases response times, hinders the allocation of resources, and limits the operational capacity of the institutions that administer justice. Public prosecution services are exposed to it at the very start of the criminal process: every complaint they receive must be registered, assessed and processed, and a backlog at this stage propagates to the rest of the system. Anticipating where congestion is likely to arise would allow institutions to reinforce staff or redistribute workload before delays accumulate.

Machine learning has been applied to the prediction of judicial outcomes and workload, from court decisions (Aletras, Tsarapatsanis, Preoțiuc-Pietro and Lampos, 2016; Katz, Bommarito and Blackman, 2017; Medvedeva, Vols and Wieling, 2019) to case duration and congestion (Azaria, Ronen and Shamir, 2024; Vasconcelos, Sátiro, Fávero, Bortoloto and Corrêa, 2023; Pernici, Bono, Piro, Del Treste and Vecchi, 2024), and to decisions taken within the criminal process (Kleinberg, Lakkaraju, Leskovec, Ludwig and Mullainathan, 2017). These studies, however, address the court stage. Prosecutorial workload has received far less attention, and the studies on congestion rarely report a temporal evaluation in which models are selected on one period and tested on later ones, an omission that tends to produce optimistic estimates when the data drift over time (Tashman, 2000; Kapoor and Narayanan, 2023).

This study estimates a proxy risk of prosecutorial congestion from the open administrative records of the Public Prosecutor's Office of Peru (2019–2026). Its contributions are as follows. (i) It addresses congestion at the prosecutorial stage, which previous predictive studies have rarely examined, using national open records and an evaluation on a hold-out year and on a prospective partial year. (ii) It specifies a leakage-controlled protocol for a target that is derived from the same records as the predictors: the variables that define the target are excluded, and every threshold and preprocessing parameter is learned on the training years only. (iii) It shows, through the agreement between an ablation study and SHAP attributions, that the risk signal is mainly structural—carried by territorial and institutional features rather than by lagged workload—and draws the operational consequences of that finding. (iv) It reports the limits of the model explicitly, including its calibration and the sensitivity of the proxy target to its thresholds, so that its use can be delimited before any deployment.

The remainder of this paper is organized as follows. Section 2 reviews the related work. Section 3 describes the data and the methodology. Section 4 presents the results, Section 5 discusses them, Section 6 states the limitations, and Section 7 concludes.

2. Related work

2.1. Machine learning for judicial and prosecutorial workload

Prediction in the justice domain has focused mainly on the outcome of court cases, using the text of decisions or structured case features (Aletras et al., 2016; Katz et al., 2017; Medvedeva et al., 2019). A second line of work models the functioning of the system itself: case duration, productivity and congestion. Recent studies have used administrative records to reduce judicial congestion, evaluate productivity and analyze the temporal course of proceedings, combining tree-based models with explainability techniques or with process mining to locate bottlenecks (Azaria et al., 2024; Vasconcelos et al., 2023; Pernici et al., 2024). Reviews of artificial intelligence in judicial management describe a shift from isolated predictive models toward tools that support the planning of the whole system (Alhalalmeh and Al-Tarawneh, 2025).

Two gaps remain. These studies address courts, after the prosecutorial investigation stage, and their evaluation designs seldom separate the period used to build the model from the period used to test it. The present study addresses the prosecutorial stage and adopts such a separation.

2.2. Feature selection: filter, wrapper, embedded, and ensemble approaches

Feature selection affects the stability, interpretability and generalization of classifiers (Saeys, Abeel and Van de Peer, 2008). Filter methods rank features by statistical association with the target; wrapper methods search subsets using the performance of a classifier, at a higher computational cost (Zhou, Yuan, Gao and Yang, 2024); and embedded methods select features while the model is trained, so their result depends on the algorithm. All-relevant selectors such as Boruta compare the importance of each feature with randomized shadow copies, which yields a relevance test rather than a minimal subset (Kursa and Rudnicki, 2010).

Individual selectors often disagree when features are correlated, which makes the selected subset depend on the algorithm. Ensemble or consensus selection aggregates several criteria to make the subset more stable (Saeys et al., 2008; Bolón-Canedo and Alonso-Betanzos, 2019; Moslemi, 2023; Drotár, Gazda and Vokorokos, 2018). Following this approach, the present study combines six methods—two filters for continuous variables (mutual information and ANOVA), one filter for categorical variables (χ^2), one embedded method (random forest importance (Breiman, 2001)), and two wrappers (Boruta and RFECV, the cross-validated form of recursive feature elimination (Guyon, Weston, Barnhill and Vapnik, 2002))—after an explicit multicollinearity audit with Pearson's correlation and Cramér's V.

2.3. Class imbalance, data leakage, and temporal validation

Risk classification on administrative data is usually imbalanced. Synthetic oversampling such as SMOTE (Chawla, Bowyer, Hall and Kegelmeyer, 2002) is a common response; class weighting is a simpler alternative that leaves the data unchanged, and it is the one adopted here (Section 3.4.2), since the observed imbalance (86%/14%) leaves enough positive cases for training.

Data leakage—the use, during model development, of information that would not be available when the model is applied—is a frequent source of optimistic results (Kaufman, Rosset, Perlich and Stitelman, 2012). A review across disciplines found that it affects hundreds of published machine learning studies, often because preprocessing statistics are computed before the data are split (Kapoor and Narayanan, 2023). Restricting preprocessing to the training data and validating on later periods are the recommended remedies (Sasse, Nicolaisen-Sobesky, Dukart, Eickhoff, Götz, Hamdan, Komeyer, Kulkarni, Lahnakoski, Love, Raimondo and Patil, 2023; Hamdan, More, Sasse, Komeyer, Patil and Raimondo, 2023). When observations are ordered in time, random cross-validation mixes past and future; forecasting research therefore recommends evaluating on blocks of data that follow the training period (Tashman, 2000; Bergmeir and Benítez, 2012; Roberts, Bahn, Ciuti, Boyce, Elith, Guillera-Aroita, Hauenstein, Lahoz-Monfort, Schröder, Thuiller, Warton, Wintle, Hartig and Dormann, 2017), and the literature on concept drift and dataset shift explains why performance can change from one period to the next (Gama, Žliobaitė, Bifet, Pechenizkiy and Bouchachia, 2014; Moreno-Torres, Raeder, Alaiz-Rodríguez, Chawla and Herrera, 2012).

2.4. Explainability and responsible AI for risk prediction in judicial contexts

Interpretability is widely regarded as a requirement for the adoption of predictive models in judicial contexts, where explanations must be understood by different institutional stakeholders (Richmond, Muddamsetty, Gammeltoft-Hansen, Olsen and Moeslund, 2024). For high-stakes decisions, some authors argue for inherently interpretable models instead of post hoc explanations of black boxes (Rudin, 2019); others use additive attributions such as SHAP, which decompose each prediction into feature contributions and have exact, efficient forms for tree ensembles (Lundberg and Lee, 2017; Lundberg, Erion, Chen, DeGrave, Prutkin, Nair, Katz, Himmelfarb, Bansal and Lee, 2020; Saarela and Podgorelec, 2024; Bhatnagar and Huchhanavar, 2025). Risk assessment tools in criminal justice have also been scrutinized for their accuracy and fairness: a widely used commercial tool was no more accurate than predictions made by people without legal training (Dressel and Farid, 2018), and fairness criteria for such tools can be mutually incompatible (Berk, Heidari, Jabbari, Kearns and Roth, 2021). High accuracy alone is therefore insufficient; institutional trust also requires transparent and auditable models (Taylor, 2025).

The present study uses SHAP attributions together with an ablation study, so that conclusions about which features matter rest on two independent sources, and it reports the calibration of the scores before discussing their operational use.

3. Materials and methods

3.1. Dataset

3.1.1. Description

This study uses the administrative records from the "MPFN Fiscal Cases" dataset, published by the Public Prosecutor's Office – Office of the Attorney General of Peru (MPFN) on the National Open Data Platform (Ministerio Público – Fiscalía de la Nación (MPFN), 2022). The dataset reports the volume of cases received and processed nationwide, disaggregated by prosecutorial district, prosecution office type, subject matter, and specialty, and distributed in independent annual CSV files covering the 2019–2026 period.

3.1.2. Variables

The official MPFN data dictionary comprises 16 raw administrative fields structured into four dimensions: temporality, institutional geography, case taxonomy, and volumetric workload, as described in Table 1.

Table 1

Raw administrative fields from the MPFN data dictionary. `ubigeo_pjfs` is encoded according to the INEI catalog.

Dimension	Variables
Temporality	<code>periodo</code> , <code>anio</code> , <code>fecha_descarga</code> , <code>fecha_corte</code>
Institutional Geography	<code>distrito_fiscal</code> , <code>ubigeo_pjfs*</code> , <code>dpto_pjfs</code> , <code>prov_pjfs</code> , <code>dist_pjfs</code>
Case Taxonomy	<code>tipo_fiscalia</code> , <code>materia</code> , <code>especialidad</code> , <code>tipo_caso</code> , <code>especializada</code>
Volumetric Workload	<code>ingresado</code> , <code>atendido</code>

Based on these fields, 14 variables were constructed through feature engineering and grouped into four families: operational pressure indicators, temporal descriptors, second-order categorical interactions, and lagged historical metrics by prosecutorial district, computed as moving averages and year-over-year growth rates of case inflows and processed cases from the previous period (Table 2).

3.1.3. Study period

The unified dataset covers the period from January 2019 to May 2026 (partial), consolidating a matrix of 9,593 records after the integration process. Because the records are ordered in time, the data were partitioned by calendar year rather than at random, following the recommendations for out-of-sample evaluation of time-ordered data (Tashman, 2000; Bergmeir and Benítez, 2012).

The chronological distribution was configured as follows: the 2019–2023 period was reserved for the primary training stage (subdivided into an internal training block spanning 2019–2022 and an internal validation block

Table 2

Variables derived through feature engineering. [†] Subsequently excluded from the candidate feature matrix due to direct collinearity with the calendar year (Section 3.2.5).

Dimension	Variables
Operational Pressure	saldo_casos, tasa_atencion, ratio_saldo
Temporal Descriptors	anio_centrado [†] , post_pandemia, periodo_pandemia_2020
Categorical Interactions (2nd Order)	inter_distrito_tipo_caso, inter_materia_tipo_fiscalia, inter_tipo_fiscalia_especialidad
Lagged Historical Metrics (by Prosecutorial District)	hist_ingresado_mean_prev_dist_pjfs, hist_atendido_mean_prev_dist_pjfs, hist_saldo_mean_prev_dist_pjfs, growth_ingresado_prev_dist_pjfs, growth_atendido_prev_dist_pjfs

corresponding to 2023, used for consensus-based feature selection); the 2024 block was allocated to model selection, threshold selection and probabilistic calibration; the 2025 observations were isolated as the hold-out test set; finally, the available records from 2026 were used as an independent prospective evaluation set. A summary of these characteristics is presented in Table 3.

Table 3

General characteristics of the dataset. The partition is reported as train/validation/test/external. Class balance is expressed as Class 0 / Class 1.

Characteristic	Value
Source	MPFN – Peru Open Data [@ministerio_publico__fiscalia_de_la_nacion_mpfm_mpfm_2022]
Period	Jan. 2019 – May 2026
Total Records	9,593
Original Fields	16
Candidate Features (Initial)	28
After Pearson Filter (>0.90)	26
After Cramér's V Filter (>0.80)	17
After Encoding (One-Hot + Frequency)	433
Final Features (6-Method Consensus)	69
Train/Valid/Test/External Split	6076 / 1195 / 1199 / 1123
Class Balance (Train)	86.0% / 14.0%
Validation	Year-block, 4 folds

3.2. Methodological framework

The methodology consists of six sequential phases, from data quality to feature selection, designed so that every step can be audited and repeated (Fig. 1).

3.2.1. Data quality

This phase included the normalization of categorical variables through the removal of leading and trailing whitespace, standardization to uppercase, and harmonization of missing values, as well as the detection of redundant columns resulting from the annual data integration process and the removal of duplicate records.

For the missing values identified in the *ingresado* and *atendido* fields, imputation was performed using the median computed exclusively from the temporal training blocks to avoid data leakage (Kapoor and Narayanan, 2023). In parallel, binary indicator variables (*flag_nulo_*) were generated to preserve the missingness signal as a potentially predictive attribute.

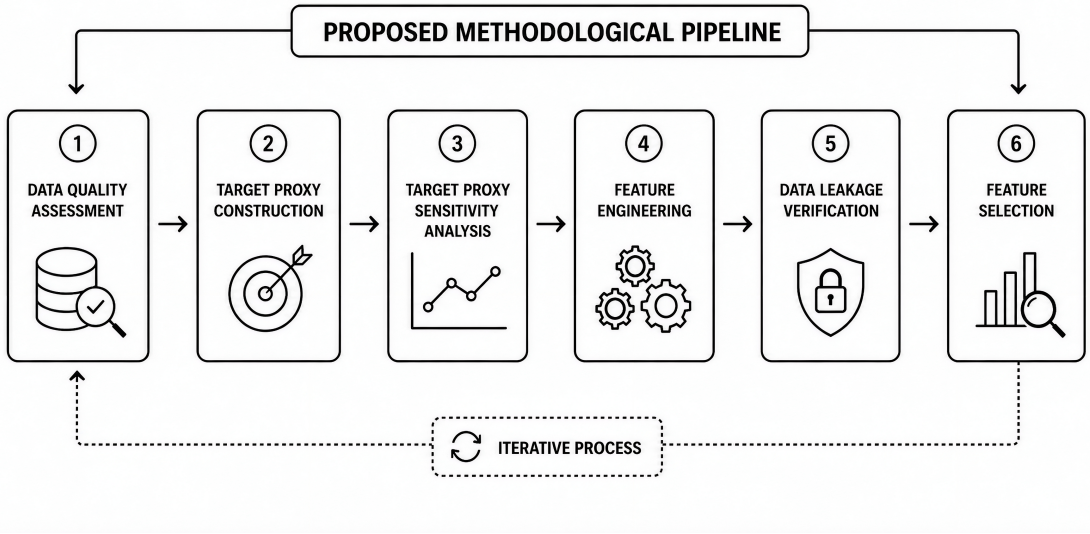


Figure 1: Methodological workflow: from data quality assessment to consensus feature selection.

For categorical variables, missing values were imputed using an explicit category (*NO_ESPECIFICADO*) instead of the mode, thereby avoiding the introduction of an artificially dominant class. Finally, the interquartile range (IQR) criterion was applied for the preliminary profiling and identification of numerical outliers.

3.2.2. Proxy target

Let $s_i = \text{ingresado}_i - \text{atendido}_i$ denote the case backlog and $t_i = \text{atendido}_i / \text{ingresado}_i$ the case processing rate for record i . The binary target variable *riesgo_congestion* is defined, under the baseline scenario (P75/P25), as shown in Eq. 1.

$$y_i = \mathbb{1} [s_i \geq q_{.75}(s_{\text{train}}) \wedge t_i \leq q_{.25}(t_{\text{train}})] \quad (1)$$

where the percentile thresholds $q_{.75}$ and $q_{.25}$ are computed exclusively from the training block. The label is an operational proxy signal, not an official certification of prosecutorial congestion nor a causal quantity. It follows a conservative criterion: a record is flagged only when a high backlog and a low processing rate occur together.

3.2.3. Target sensitivity analysis

Since *riesgo_congestion* is a proxy label, its robustness to the choice of percentile thresholds was examined before it was adopted as the basis for modeling. To this end, three alternative scenarios were established as a parallel sensitivity analysis (Table 4).

Table 4

Sensitivity analysis of the proxy target across scenarios. Train % = prevalence in the training set; Test % = prevalence in the 2025 test set; Ext % = prevalence in the 2026 external dataset; CV = interannual coefficient of variation (2019–2026); Role = operational interpretation of the scenario.

Threshold	Train %	Test %	Ext %	CV	Role
P70/P30	19.4	19.8	26.3	0.165 (min.)	Early warning
P75/P25	14.0	14.4	20.2	0.219	Primary
P80/P20	9.3	9.0	14.3	0.278	Strict
P85/P15	5.3	5.0	9.3	0.379 (max.)	Very strict

The prevalence of Class 1 in the training set ranges from 5.3% (P85/P15) to 19.4% (P70/P30), while the temporal instability of the event rate—measured as the coefficient of variation (CV) between 2019 and 2026—increases

monotonically with threshold severity (CV = 0.165 for P70/P30 up to 0.379 for P85/P15). The baseline scenario (P75/P25, CV = 0.219, training prevalence = 14.0%) is not the most stable of the four—P70/P30 is—but it was retained as a compromise between institutional severity and a class balance sufficient for training without synthetic resampling, avoiding the greater rarity and interannual variability of the P80/P20 and P85/P15 scenarios. The prevalence also rises in the 2026 data under every scenario (Table 4), a shift that the evaluation on that year must take into account.

3.2.4. Feature engineering

The 14 derived variables were designed according to four criteria: operational pressure indicators (imbalance between case inflows and processed cases), pandemic-related temporal descriptors (*post_pandemia*, *periodo_pandemia_2020*), second-order categorical interactions that capture nonlinear institutional synergies, and lagged historical variables by prosecutorial district that use only previous periods, so that no information from the year being predicted enters the features.

3.2.5. Data leakage control

Because the target variable (*riesgo_congestion*) is constructed directly from *ingresado* and *atendido*, these attributes—*ingresado*, *atendido*, and their derived variables *saldo_casos*, *tasa_atencion*, and *ratio_saldo* (LEAK-AGE_COLS)—were removed through a strict exclusion list, together with non-predictive metadata and the columns corresponding to the four alternative proxy scenarios. Without this exclusion the target could be reconstructed from the predictors, a typical form of leakage (Kaufman et al., 2012). Additionally, *anio_centrado* was excluded from the candidate feature matrix because of its direct collinearity with the calendar year (*COLLINEAR_EXCLUDE*).

3.2.6. Feature selection

A multicollinearity audit was first performed using the Pearson correlation coefficient ($r > 0.90$) for numerical variables and Cramér's V (> 0.80) for categorical variables; the number of candidate features retained after each filter is given in Table 3.

Subsequently, six complementary feature selection methods—two continuous filter methods, one categorical filter method, one embedded method, and two wrapper methods (Table 5)—were applied, each operating under the same year-block temporal cross-validation scheme.

Table 5
Feature selection methods evaluated.

Method	Type	No. vars
Mutual Information	Filter (Continuous)	40
ANOVA F-test	Filter (Continuous)	40
χ^2	Filter (Categorical)	40
Random Forest	Embedded	40
Boruta	Wrapper (All-Relevant)	12
RFECV	Wrapper (Recursive + Temporal CV)	52

The consensus among methods is formalized by Eq. 2, where each term represents the binary vote of the corresponding method (1 if feature i is selected, 0 otherwise).

$$FeatureScore_i = \frac{MI_i + ANOVA_i + \chi_i^2 + RF_i + Boruta_i + RFECV_i}{6} \quad (2)$$

The final features were determined using a consensus voting mechanism with a minimum threshold of 2 out of 6 methods, resulting in a subset of 69 final features (Table 6). A low threshold was chosen deliberately: it keeps any feature that two methods of different families consider relevant, and leaves the final weighting to the classifiers.

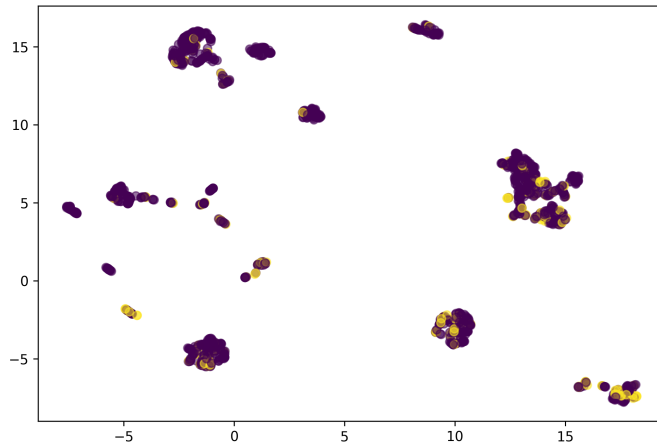
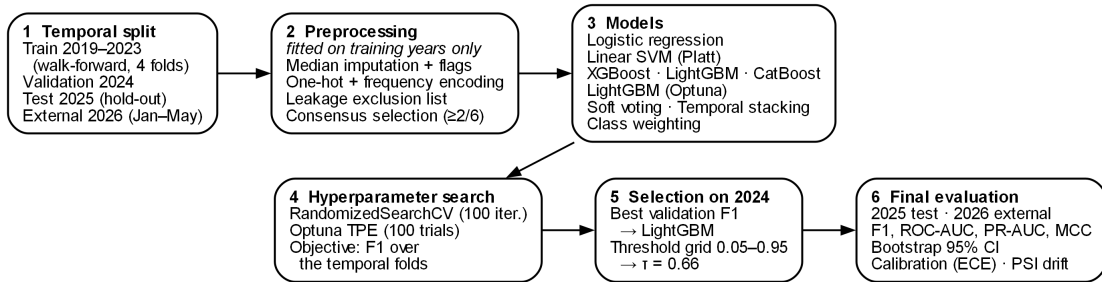
3.3. Feature space validation

Before training, the structure of the training feature space was examined to check whether the proxy risk is consistent with its multivariate organization. Two-dimensional projections (PCA, t-SNE and UMAP) showed nonlinear structure with local clusters and partial overlap between classes (Fig. 2 shows the UMAP projection), which motivated the use

Table 6

Consensus voting distribution and retained features. Consensus threshold = votes $\geq 2/6$. The three features with unanimous agreement (6/6) are freq_distrito_fiscal, freq_tipo_caso, and tipo_caso_DENUNCIA.

Votes (6 methods)	No. vars	% Total (433)
6/6 (Unanimous)	3	0.7%
5/6	5	1.2%
4/6	12	2.8%
3/6	20	4.6%
2/6 (Minimum Threshold)	29	6.7%
Retained Subtotal (≥ 2 Votes)	69	15.9%
1/6 (Discarded)	15	3.5%
0/6 (Discarded)	349	80.6%

**Figure 2:** Two-dimensional UMAP representation of the feature space of the training set.**Figure 3:** Experimental design for model training and temporal validation.

of nonlinear classifiers. The unsupervised validation (Table 7) found that K-Means ($k = 7$) gave the partition with the highest quality and stability among the clustering methods compared, and that anomaly detectors flagged only a small share of the records.

3.4. Experimental design

The experimental design is summarized in Fig. 3 and described in the rest of this section.

Table 7

Summary of the unsupervised validation of the feature space.

Analysis	Evidence	Result + Interpretation
PCA (2 Components)	2D projection	Global organization; partial nonlinear separation
t-SNE	2D projection	Compact local clusters; nonlinear relationships
UMAP	Fig. 2	Stable structure; consistent multivariate patterns
K-Means	Silh = 0.262; DB = 1.507; CH = 1173.6; ARI = 0.77 ± 0.13	Optimal (K = 7); robustness confirmed by bootstrap
DBSCAN	46 clusters; Silh = 0.161; DB = 1.310; CH = 245.9	Exploratory; fragmented partition with lower quality
Agglomerative Clustering	Silh = 0.250 ($k = 5$); DB = 1.573; CH = 1273.8	Exploratory; lower quality than K-Means
Isolation Forest	51/6076 (0.84%)	Low proportion; no large-scale anomalies
Local Outlier Factor	54/6076 (0.89%)	Consistent with the training set
Robust Mahalanobis Distance	152/6076 (2.50%)	No extreme outliers

3.4.1. Temporal data partitioning

The experimental design replaces random cross-validation with a cumulative chronological strategy based on annual blocks (year-block walk-forward validation), so that every model is evaluated on years that follow its training data (Bergmeir and Benítez, 2012; Roberts et al., 2017). Within the primary training block (2019–2023), sequential folds are organized chronologically; at each iteration, the model is trained exclusively on periods preceding the year under internal evaluation.

Externally, the 2024 period was isolated as the validation and calibration set (for classifier selection and threshold determination); the 2025 observations were reserved as the final hold-out test set; and the partial 2026 segment was used as an independent prospective evaluation set. None of these three temporal horizons participated in the hyperparameter optimization process.

3.4.2. Model configurations

Nine algorithmic configurations were evaluated (Table 8): one baseline classifier, two linear models (logistic regression and a linear support vector machine whose scores were calibrated with Platt scaling through *Calibrated-ClassifierCV* (Niculescu-Mizil and Caruana, 2005)), three gradient boosting implementations—XGBoost (Chen and Guestrin, 2016), LightGBM (Ke, Meng, Finley, Wang, Chen, Ma, Ye and Liu, 2017) and CatBoost (Prokhorenkova, Gusev, Vorobev, Dorogush and Gulin, 2018)—one Bayesian-optimized LightGBM variant, and two ensemble meta-models. Gradient boosting was included because it handles mixed, nonlinear and interacting features without manual specification, which the feature space analysis suggested (Section 3.3).

To account for the class imbalance of the dataset (86% versus 14%), compatible models used class weighting in their loss functions (*class_weight* = "balanced" or *scale_pos_weight*); synthetic resampling was not used, so the training data remained the observed records.

Two meta-models were then built from the five base estimators with the highest internal cross-validation F1 scores—logistic regression, LightGBM, Optuna-optimized LightGBM, XGBoost and CatBoost, excluding the SVM because of its lower performance (CV F1 = 0.268): a soft voting ensemble (*VotingClassifier*), which averages the predicted probabilities of the five models (Eq. 3), and a temporal stacking classifier (Wolpert, 1992), which trains a logistic regression on the five out-of-fold probabilities (Eq. 4).

$$\hat{p}(y = 1 | x) = \frac{1}{5} \sum_{k=1}^5 \hat{p}_k(y = 1 | x) \quad (3)$$

Table 8

Model configurations evaluated (hyperparameter search). F1 = mean F1-score over the internal temporal folds. The complete hyperparameter search spaces will be released with the code repository upon publication.

Model	Optimizer	No. of Iterations	F1
Dummy	—	—	—
Logistic Regression	RandomizedSearchCV	100	0.485
SVM (LinearSVC + Platt)	RandomizedSearchCV	100	0.268
XGBoost	RandomizedSearchCV	100	0.553
LightGBM	RandomizedSearchCV	100	0.563
CatBoost	RandomizedSearchCV	100	0.563
LightGBM (Optuna)	TPE (Optuna)	100	0.568
VotingClassifier	— (Ensemble, No Hyperparameter Tuning)	—	—
Temporal Stacking	— (Meta-Model Based on OOF Predictions)	—	—

3.4.3. Optimization strategy

The hyperparameters of the base estimators were tuned using *RandomizedSearchCV* (N = 100 iterations), which explores large search spaces more efficiently than grid search (Bergstra and Bengio, 2012), maximizing the F1-score over the temporal folds defined previously.

The LightGBM architecture was also explored using Bayesian optimization (TPE in *Optuna* (Akiba, Sano, Yanase, Ohta and Koyama, 2019), 100 trials) as a comparative configuration (Table 8); the final selected model was LightGBM optimized with *RandomizedSearchCV*, as it achieved a higher F1-score on the 2024 validation set (Table 9).

The stacking meta-model (logistic regression) is trained on the out-of-fold probabilities generated by the five base models (Eq. 4).

$$\hat{p}_{\text{stack}}(y = 1 \mid x_i) = \sigma(w^T z_i + b), \quad z_i = [\hat{p}_1(x_i), \dots, \hat{p}_5(x_i)] \quad (4)$$

where σ denotes the logistic function, and w and b are the parameters estimated by logistic regression. The meta-model was trained exclusively on the out-of-fold predictions of the five base models across the four temporal folds, so that it never sees a prediction made on a model's own training data. For inference, the base models were retrained on the entire training set while keeping the meta-model fixed.

3.4.4. Evaluation protocol

The primary classifier was selected by comparing the F1-score and ROC-AUC on the 2024 validation block, a criterion fixed before the 2025 test set was evaluated; LightGBM was selected.

The decision threshold was chosen through a grid search (91 values within [0.05, 0.95]) that maximizes the F1-score on the validation set, yielding $\tau = 0.66$. Alternative thresholds (Youden's J index, target precision and target recall) were also computed for operational sensitivity analysis.

The final performance was reported on the 2025 test set and on the 2026 prospective block using accuracy, precision, recall, F1-score, ROC-AUC, PR-AUC, balanced accuracy, specificity, the Matthews correlation coefficient (MCC), Cohen's kappa, negative predictive value (NPV), false positive rate (FPR), false negative rate (FNR), and likelihood ratios (LR+ and LR−). PR-AUC and MCC were included because, under class imbalance, they are more informative than accuracy or ROC-AUC alone (Saito and Rehmsmeier, 2015; Chicco and Jurman, 2020). Differences in ROC-AUC between models were tested with DeLong's test (DeLong, DeLong and Clarke-Pearson, 1988), and differences across the temporal folds with the Friedman test (Demšar, 2006).

The estimates were complemented with 95% confidence intervals obtained through bootstrap resampling, calibration curves, and temporal drift analysis of the features using the Population Stability Index (PSI) with respect to the training block.

4. Results

4.1. Comparison of base models and ensembles

Table 9 summarizes the performance of the base models and the two ensemble methods (*VotingClassifier* and temporal *Stacking*) on the 2024 validation set and the 2025 test set, using a fixed decision threshold of 0.5.

Table 9
Model comparison (threshold = 0.5).

Model	Validation F1	Test F1	Test ROC-AUC
LightGBM (Randomized SearchCV)	0.605	0.475	0.798
CatBoost	0.591	0.485	0.817
XGBoost	0.598	0.465	0.772
Logistic Regression	0.437	0.444	0.799
SVM (LinearSVC + Platt)	0.263	0.247	0.803
VotingClassifier	0.584	0.476	0.819
Temporal Stacking	0.518	0.445	0.822
Dummy (Baseline)	0.000	0.000	0.500

The gradient boosting models obtained higher validation F1-scores than the linear models and the baseline classifier (F1 = 0 across all partitions), which is consistent with a nonlinear predictive signal. LightGBM achieved the highest F1-score on the validation set—the criterion used to select the primary model—whereas CatBoost obtained a higher F1-score (0.485 versus 0.475) and a higher ROC-AUC (0.817 versus 0.798) on the 2025 test set. DeLong's test indicated that the difference in ROC-AUC was statistically significant ($z = 2.70$, $p = 0.007$); the test does not apply to the threshold-dependent F1-score. The Friedman test across the temporal folds indicated overall differences among the evaluated models ($\chi^2 = 12.6$, $p = 0.013$), although with four folds its power to rank individual models is limited (Demšar, 2006).

Selecting CatBoost on the basis of its 2025 results would have turned the hold-out year into a model selection criterion, breaking the temporal isolation established in Section 3.4.1. LightGBM was therefore retained as the primary model, as fixed by the pre-specified 2024 criterion (Section 3.4.4). It also showed slightly greater stability across the walk-forward evaluation (mean F1 = 0.541 ± 0.045 versus 0.528 ± 0.040 for CatBoost; Section 4.2).

4.2. Decision threshold optimization and final model performance

With the threshold optimized on the validation set ($\tau = 0.66$; Section 3.4.4), the final model reached an F1-score of 0.618 on the validation set, 0.469 on the 2025 test set (95% bootstrap confidence interval [0.390, 0.546], 300 resamples) and 0.559 on the 2026 external data (Table 10). Specificity remained above 0.92 in all three periods, whereas recall dropped from 0.667 in 2024 to 0.424 in 2025.

Table 10
Performance of the final model (LightGBM, threshold = 0.66) across temporal partitions.

Metric	Validation 2024	Test 2025	External 2026
Accuracy	0.890	0.862	0.842
Precision	0.576	0.525	0.644
Recall	0.667	0.424	0.493
F1-score	0.618	0.469	0.559
ROC-AUC	0.897	0.798	0.866
PR-AUC	0.625	0.499	0.652
Matthews Correlation Coefficient (MCC)	0.556	0.394	0.471
Specificity	0.925	0.936	0.931

The drop from validation to test is expected under temporal drift (Gama et al., 2014), and the higher precision and PR-AUC in 2026 coincide with the higher prevalence of the target in that year (Table 4), to which precision-based metrics are sensitive (Saito and Rehmsmeier, 2015). External walk-forward validation (retraining the model year by year) gave a mean F1 of 0.541 ± 0.045 for LightGBM, 0.528 ± 0.040 for CatBoost and 0.461 ± 0.025 for logistic regression across seven temporal evaluation points (2020–2026).

4.3. Ablation study of feature groups

Table 11 summarizes the ablation study conducted on the LightGBM model, which removes entire feature families and measures the effect on the 2025 test set at the fixed threshold of 0.5.

Table 11
Ablation study (LightGBM, 2025 test set, threshold = 0.5).

Variant	No. vars	F1	Recall	Precision	ROC-AUC	PR-AUC
Full Model (69 Features)	69	0.466	0.541	0.410	0.788	0.507
Without Historical Growth Features	66	0.471	0.552	0.411	0.810	0.521
Without Frequency Encoding	64	0.458	0.622	0.363	0.818	0.522
Without Interaction Features	26	0.415	0.634	0.309	0.778	0.348
Without Territorial Features	23	0.415	0.721	0.292	0.792	0.367

Removing the categorical interaction features or the territorial attributes produced the largest decreases in F1-score (0.415 in both cases, compared with 0.466 for the full model). Without the territorial features the model became less selective—recall rose from 0.541 to 0.721 while precision fell from 0.410 to 0.292—and without the interaction features PR-AUC fell from 0.507 to 0.348, the largest decrease observed. ROC-AUC, in contrast, changed little in both variants (0.792 and 0.778 versus 0.788), so the ranking of records was largely preserved and much of the loss in F1-score reflects the fixed threshold of 0.5 rather than a loss of discrimination.

Removing the historical growth features did not degrade performance (F1 = 0.471), and removing frequency encoding changed the F1-score only marginally (0.458) while ROC-AUC rose to 0.818. Both results point to redundancy between these groups and the territorial interaction features, consistent with the multicollinearity audit.

4.4. Global feature importance

Permutation importance (Altmann, Tološi, Sander and Lengauer, 2010) and the mean absolute SHAP values (Section 5) identify the same leading predictors: the territorial-institutional frequency variables and the *tipo_caso_DENUNCIA* feature, in agreement with the ablation study. The SHAP interaction analysis identified *frec_inter_distrito_tipo_caso* $\times$ *ubigeo_pifs* as the feature pair with the highest mean interaction value (0.108). A controlled counterfactual analysis showed that, in several individual cases, changing a single feature was sufficient to cross the decision threshold, and that these features were institutional descriptors of the case rather than measures of accumulated workload.

4.5. Calibration and consistency with the feature space

Fig. 4 shows that the model is reasonably calibrated for low predicted probabilities but overconfident in the medium-to-high range (ECE = 0.091; MCE = 0.464). Because the operational threshold ($\tau = 0.66$) lies in that range, the score should be read as a ranking criterion for prioritization rather than as a calibrated probability.

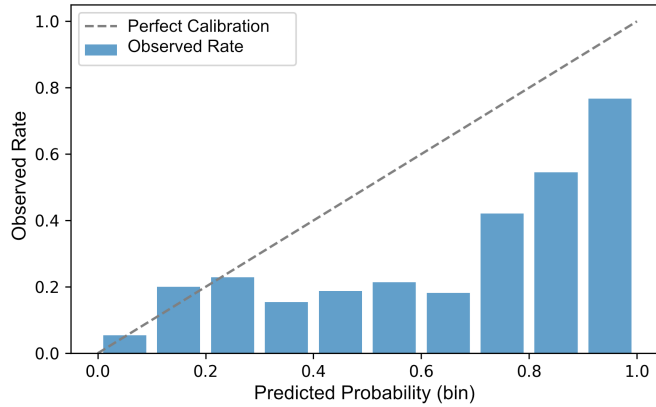


Figure 4: Reliability diagram of the final model (LightGBM, $\tau = 0.66$) on the 2025 test set.

The clusters of the unsupervised analysis with the highest average case backlog also concentrated the highest proxy risk rates (Table 7), so the structure found without labels is consistent with the one the classifier learned.

5. Discussion

5.1. Interpretability

The interpretability analysis uses SHAP (Lundberg and Lee, 2017; Lundberg et al., 2020), applied to the final model on the 2025 test set. Fig. 5 presents the mean absolute SHAP values of the most influential features. The model bases its predictions mainly on territorial and institutional patterns: features describing territorial organization, institutional specialization and operational frequency carry the largest contributions, which suggests that the proxy risk reflects specific operational configurations of the MPFN rather than the volume handled by each combination. Five features account for 53% of the total SHAP signal, whereas the 50 least influential features contribute 6%. The historical features play a secondary role, consistent with the ablation study (Table 11).

Fig. 6 shows how the contribution of territorial frequency varies with geographic location across individual predictions. Some features contribute in a stable way, whereas others vary with the operational context, a sign of the interactions that a nonlinear model can represent (Lundberg et al., 2020). The risk therefore does not depend on location or case type in isolation but on their combination, which agrees with the ablation result that removing the interaction features degrades performance more than removing any other group. These findings are predictive associations learned by the model, not causal relationships.

5.2. A structural signal

The dominance of territorial features over historical ones follows from the nature of the target. Because the label is defined with percentiles of backlog and processing rate, it captures persistent overload associated with the installed capacity of each prosecutorial district and office type, rather than year-to-year dynamics. This has two consequences. On the one hand, it explains why the ablation penalizes the removal of the territorial interaction features more than the removal of the trend features. On the other, it means that a large part of what the model predicts is which operational combinations tend to be congested, a property that is partly persistent over time. The value of the model, then, lies less in anticipating sudden changes than in ranking combinations consistently with the information available before each year, under a leakage-free protocol. How much of this ranking would be recovered by simply carrying forward the previous year's label is an open question that the present design does not answer (Section 6).

5.3. Model behavior

The stability of LightGBM relative to CatBoost may be related to the ordered categorical encoding of the latter, which can respond more strongly to shifts in category frequencies across years (Prokhorenkova et al., 2018); such shifts were monitored with the PSI (Section 3.4.4). CatBoost's advantage in 2025 should therefore be read as a property of

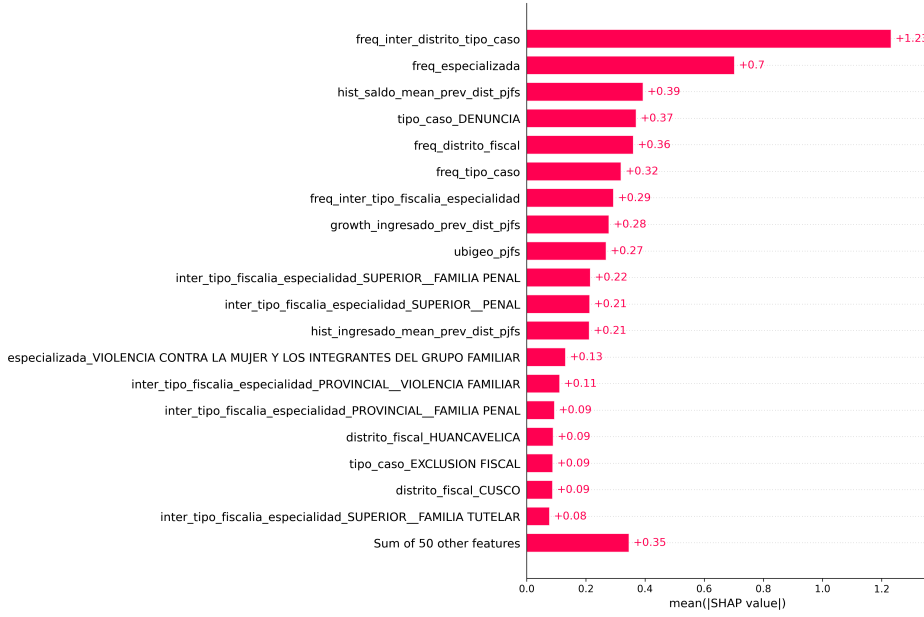


Figure 5: Global feature importance according to mean SHAP values.

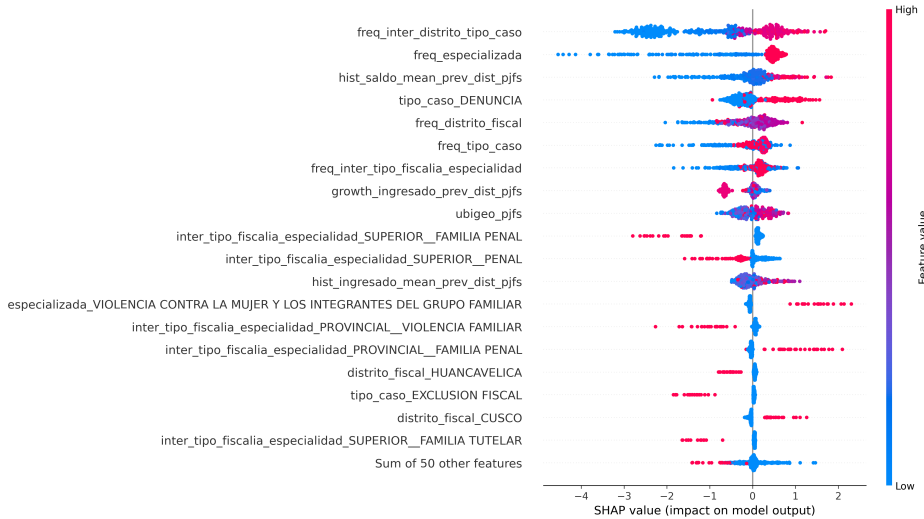


Figure 6: SHAP interaction between territorial frequency and geographic location.

that partition rather than as systematic superiority. The limited gain of the stacking model is explained by the high correlation among the base models—four of which are gradient boosting variants—which reduces the diversity a meta-model needs (Wolpert, 1992). Finally, the low F1-score of the linear SVM (0.268) agrees with the unsupervised analysis (Section 3.3): the classes are not linearly separable in the feature space.

5.4. Operational impact

To illustrate the operational consequences of the errors, a sensitivity analysis with parametric costs was conducted. A 10:1 cost ratio between a false negative and a false positive was assumed, reflecting the greater institutional impact of missing a congestion alert. Under this assumption, a cost of S/ 50,000 was assigned to each false negative and S/ 5,000 to each false positive. On the 2025 test set (99 FN, 66 FP, 73 TP and 961 TN), the model yields an illustrative

loss of S/ 5,280,000 (Table 12). These values are not official costs of the MPFN; they serve only to compare operating scenarios, such as alternative thresholds.

Table 12

Operational impact under illustrative cost assumptions. The reported costs are parametric assumptions, not audited figures.

FN	FP	TP	TN	FN Cost	FP Cost	Estimated Loss	FN:FP
99	66	73	961	S/ 50,000	S/ 5,000	S/ 5,280,000	10:1

With a recall of 0.424 in 2025, the model misses more than half of the combinations at risk, so it is suited to support, not replace, the institution's own monitoring: its scores can order the combinations to be reviewed first, while the final decision remains with practitioners. This use is consistent with studies that identify institutional organization and operational patterns as relevant factors in the predictive analysis of judicial systems (Azaria et al., 2024; Vasconcelos et al., 2023), and with the view that predictive tools in justice should be auditable and remain under human control (Rudin, 2019; Taylor, 2025).

6. Limitations

The target is a proxy of prosecutorial congestion built with percentile rules; it is not an official measure of the MPFN and has not been validated by external expert judgment, although its thresholds were examined in a sensitivity analysis. The unit of analysis is an aggregated operational combination rather than an individual case, so predictions support operational prioritization, not the assessment of specific cases.

Because the signal is mainly structural, part of it may be captured by the persistence of the label over time. The study compares the models against a non-informative baseline but not against a persistence rule that carries forward the previous year's label, nor against a model restricted to territorial identity; such comparisons are needed to quantify the added value of the learned model and are left for future work.

The 2023 block was used both for consensus feature selection and as part of the internal temporal folds, so the internal cross-validation estimates may be optimistic; the 2024, 2025 and 2026 evaluations are unaffected. The Friedman test over four folds has limited power, and the scores are overconfident in the range where the decision threshold lies, so they should not be interpreted as calibrated probabilities. The 2026 data cover only January to May and show a higher prevalence of the target, which limits the comparability of that year with the complete ones.

The results come exclusively from Peruvian data (2019–2026); applying the approach to other justice systems requires retraining and local validation. The cost analysis relies on illustrative assumptions, and the relationships identified are predictive associations, not causal effects. Finally, the small number of positive cases in some subgroups did not allow a quantitative assessment of performance across office types and specialties, which remains necessary before any institutional use (Berk et al., 2021).

7. Conclusions

This study estimated a proxy risk of prosecutorial congestion from the open records of the Public Prosecutor's Office of Peru, under a protocol that excludes the variables defining the target, learns every threshold and preprocessing parameter on the training years, selects the model and its threshold on 2024, and evaluates it on a hold-out year and on a prospective partial year. Under these conditions LightGBM reached an F1-score of 0.469 and a ROC-AUC of 0.798 on the 2025 test set and an F1-score of 0.559 on the 2026 data, with high specificity and a modest recall. These results show that a useful ranking of operational combinations can be obtained from open administrative data, while the decline from validation to test illustrates why temporal evaluation, rather than random cross-validation, is needed to estimate how such a model would perform when applied (Tashman, 2000; Kapoor and Narayanan, 2023).

The ablation study and the SHAP attributions agree that the risk is driven mainly by territorial and institutional configurations, with lagged workload adding little. Methodologically, this agreement between two independent sources strengthens the interpretation; substantively, it indicates that prosecutorial congestion, as measured here, is largely a

structural property of how offices are organized across the territory. The scores are therefore best used to prioritize the review of combinations, not as calibrated probabilities nor as a substitute for the judgment of prosecutors and managers.

Future work should compare the model with persistence and territorial-identity baselines, validate the proxy target with institutional experts, assess performance across office types and specialties, recalibrate the scores, estimate causal effects with methods such as double machine learning, and test the transferability of the approach to other justice systems as new data become available.

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